

SKILLS GUIDE



The ability to work well with crew members and the many staff who support missions on the ground.



Creativity for solving unexpected and complex problems using limited resources.



The flexibility to adapt to extreme environments and to deal with challenging living conditions.



A logical and analytical approach when handling critical and challenging situations.



The physical endurance to train for live missions, which can be hugely demanding.



An eye for detail and constant vigilance to successfully complete missions in space.



COMMANDER OR PILOT Takes responsibility for the flight of the spacecraft, as well as the safety of the crew, and the overall success of a mission. They may also perform other duties, such as helping with onboard experiments or carrying out extravehicular activity, or space walks.



FLIGHT ENGINEER Does a variety of jobs on the mission including conducting scientific experiments under micro-gravity conditions, performing routine maintenance on board space stations, and operating robotic arms to accomplish external maintenance tasks.

AT A GLANCE



YOUR INTERESTS Space • Flight • Mathematics • Physics • Mechanical engineering • Electrical engineering • Materials science



ENTRY QUALIFICATIONS As a minimum, you need at least a degree in science or engineering, or extensive experience in flying fast jets.



LIFESTYLE Working hours are irregular. Training missions involve long periods away from home, and space flights can last many months.



LOCATION Astronauts often work in remote, high-security locations, and may have to travel widely for training purposes.



THE REALITIES Working hours are long and conditions are tough and dangerous. Extensive training is both physically and mentally challenging.

RELATED CAREERS

► **MECHANICAL ENGINEER** *see pp. 182–183*

► **ELECTRICAL ENGINEER** *see pp. 186–187*

► **AIR FORCE AIRMAN** *see pp. 232–233*

► **ASTROPHYSICIST** Studies the universe using sophisticated equipment, such as satellites and telescopes. Astrophysicists study planets, stars, and other space phenomena to build and test theories about the origins and workings of the universe.

► **SATELLITE ENGINEER** Designs and builds space satellites used for relaying electronic communication, monitoring Earth, or studying the universe. Also develops scientific instruments for satellites and other associated equipment needed on the ground.